Do Now · retrieval

Data Structures · Constructors and Encapsulation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How do you create an object in C#?

Q2

What is an instance field?

Q3

What does modularity improve?

Q4

What is LIFO?

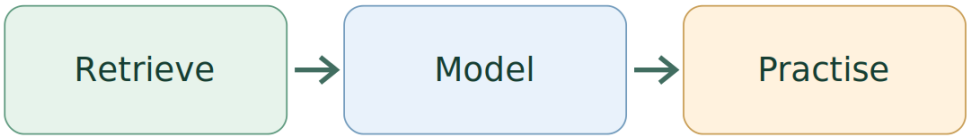
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Protect the object invariant — The constructor refuses a negative opening balance before storing it. The field is private, so outside code must use approved public methods instead of bypassing validation.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

constructor — Initialises a new object

encapsulation — Controls access to internal state

private — Accessible only within its containing type

public — Accessible from other code

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Construct a valid account; control updates

Follow the worked method

1. Constructor sets the initial state — new Account(10) invokes Account(int opening) . It checks 10 is non-negative, then stores 10 in the private field balance .

2. Method validates the proposed change — savings.Deposit(5) passes the positive-amount check and increases balance from 10 to 15. Outside code cannot change the private field directly.

3. Check valid and invalid cases — GetBalance() returns 15. Deposit(-3) throws an exception before changing the balance. Encapsulation is controlled access, not merely writing private .

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Show the value after each operation
What happens at construction, after the valid deposit, and at the attempted negative deposit?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Write a constructor for class Box that stores integer width. [4]

Question 2

Apply independently

Explain encapsulation. [2]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Suggest validation for a bank deposit method. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: a constructor has a declared return type void.

Use key vocabulary from memory

1. Explain constructor in your own words.

2. Explain encapsulation in your own words.

3. Explain private in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.